

Can Simple Trading Rules Beat Buy & Hold?

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github.com/tejasmanojav/momentum_trading

Summary

I tested whether simple trading rules can beat just buying and holding the market. I backtested moving-average, RSI, and my own "Confluence" strategy on 5 index ETFs over 20 years (2004–2024). **No strategy beat buy & hold on total return, on any of the 5 ETFs.** Buy & hold made about 791%. My best strategy, Confluence, made about 548%. What the strategies did give me was smaller crashes: the 200-day moving average's worst drop was about -24%, compared to about -55% for buy & hold. On risk-adjusted return (Sharpe ratio), they were close to a tie.

Objective

Passive index investing has always made sense to me. After reading several of John Bogle's books and watching his interviews, his main idea clicked: keep costs low and let the whole market compound without getting in the way. Reading about famous investing blunders showed me the other side. Emotional, reactive trading can eat away at wealth over time.

But I didn't want to just believe Bogle. I wanted to test it myself. Could a few simple, disciplined trading rules improve on buy & hold, or at least protect me during big crashes? To find out, I built my own backtesting tool in Python and ran it on 20 years of market data.

Hypothesis

"Can simple momentum trading strategies outperform a passive buy-and-hold approach for major indices?"

I set up my code to test three ideas:

1. Momentum strategies can reduce large drawdowns during market crashes.
2. RSI might work differently depending on how jumpy (volatile) the market is.
3. Adding more complexity and smarter risk management to the rules will make returns more consistent.

Note: I focused on index ETFs because they don't swing wildly on news or earnings from a single company. I picked five broad ETFs: SPY, QQQ, DIA, IWM, and VTI.

Data

I downloaded daily prices from Yahoo Finance using the yfinance library, on September 26, 2026. The prices are adjusted, so dividends and stock splits are included. The data starts on

January 2, 2004. Because the 200-day MA needs 200 days of history first, every strategy is measured over the same period: October 19, 2004 to December 30, 2024. When I say "average across the 5 ETFs", I mean a simple average (arithmetic mean) of the five results.

Methodology & Strategy Evolution

I didn't build everything at once. I started simple, looked for what went wrong, and fixed it one step at a time. Every strategy trades at the next day's closing price, because I only see a signal after the market closes.

1. Baseline: Buy & Hold

First I needed something to compare against. Buy & hold is pure passive investing: buy each ETF on day one and hold it for 20 years, through every crash and every rally.

2. Trend Following: Moving Average (MA)

My first strategy was the simplest trend rule there is. Buy when the price goes above its moving average, and sell when it drops below. I tried 20, 50, 100, and 200-day windows.

- **Long windows (like the 200-day)** traded less and protected against big drops, but they were slow to react.
- **Short windows (like the 20-day)** reacted fast, but they traded way too much when the market moved sideways. That meant a low win rate and lower returns.

3. Noise Reduction: MA + Buffer

In flat, sideways markets the price kept wiggling back and forth across the moving average, which meant lots of pointless trades. I needed the signal to be real before acting on it.

- **The fix:** I added a 2% buffer. The price has to move 2% past the moving average before a trade happens. I did a rough search over other buffer sizes but didn't find anything clearly better. Later, a proper test showed that 0%, 1%, and 2% all give about the same result.
- Using a percentage instead of a dollar amount means the same rule works on all five ETFs, whatever their price.

4. Risk Management: MA + Fixed Stop Loss

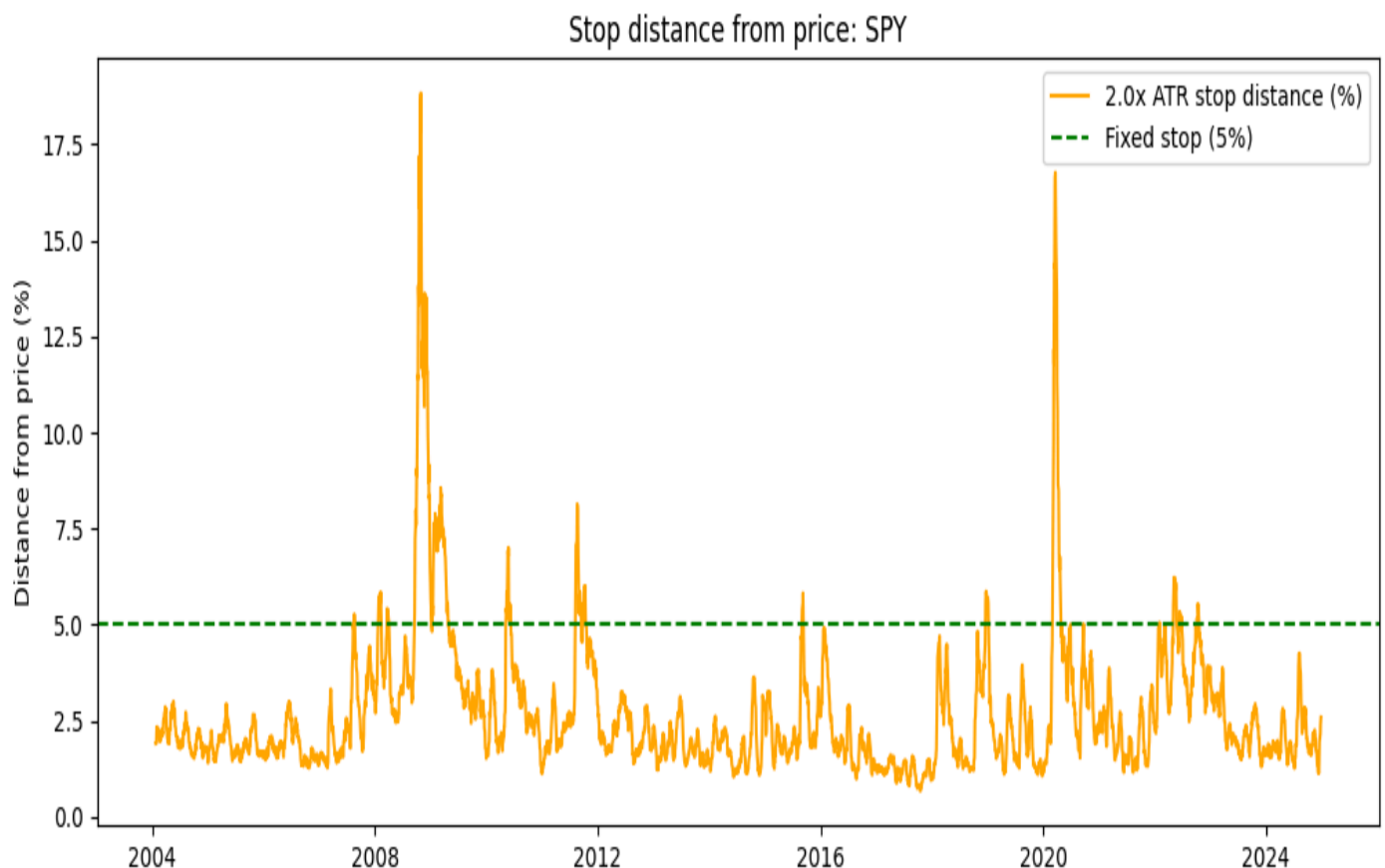
While testing, I noticed a few trades with big losses that wiped out a lot of profit. To cap my losses, I added a 5% trailing stop: if the price falls 5% from its high since I bought, I sell.

- **The trade-off:** a tight stop protects you from big drops, but it also kicks you out of good trades during normal ups and downs. That cut into my returns.

5. A Stop That Adjusts: MA + ATR Stop

Since a fixed 5% stop felt too rigid, I tried a stop that adapts to how jumpy the market is. The Average True Range (ATR) measures the typical daily price range, so an ATR stop gives the price more "room to breathe" when the market is volatile.

- **My first "winner" didn't survive.** At first, a wide 4.0x ATR stop looked like the best choice for the 200-day MA (383% vs. 330% for the fixed 5% stop). Then I fixed how trades were timed (see Technical Challenges below). After that, the top three were basically tied: ATR 3.5x at 359%, fixed 5% at 354%, and ATR 4.0x at 346%. Just moving every trade by one day reshuffled the ranking, so those small gaps are noise.
- **The 50-day MA surprise:** for the shorter 50-day window, the fixed 5% stop won clearly both before and after the fix (233% vs. 189% for ATR 3.5x after). These stop tests start as soon as each MA has enough data, so the 50-day version includes a few trades from before October 2004. The 200-day numbers aren't affected. Shorter strategies seem to do better with simple, firm exits.
- **Too tight is always bad.** A 2x ATR stop sat only about 2.7% below the price on SPY, so it got triggered all the time. It made about 4 times as many trades as the fixed 5% stop and came last every time.



6. Mean Reversion: RSI Strategy

As a different approach to trend following, I tested a mean-reversion strategy using the 14-day Relative Strength Index (RSI). The idea is to buy when an ETF looks "oversold" (RSI below 30) and sell when it looks "overbought" (RSI above 70). Early on, I used shorter 5-year windows as a quick sandbox. They were much easier to plot and look at, so I could quickly play around with different floor and ceiling levels (the RSI numbers that trigger a buy or a sell) and see what happened. In some of those windows, RSI seemed to do better. But it didn't hold up over the full 20 years, so I dropped those extra tests and kept the standard 30 and 70 levels.

Over the full 20 years, RSI won about 81% of its trades but made only about 215%. A high win rate doesn't mean high returns. RSI sells too early, sits in cash, and misses the big runs. It did best on IWM (small caps), where it made 344%, and worst on DIA (129%).

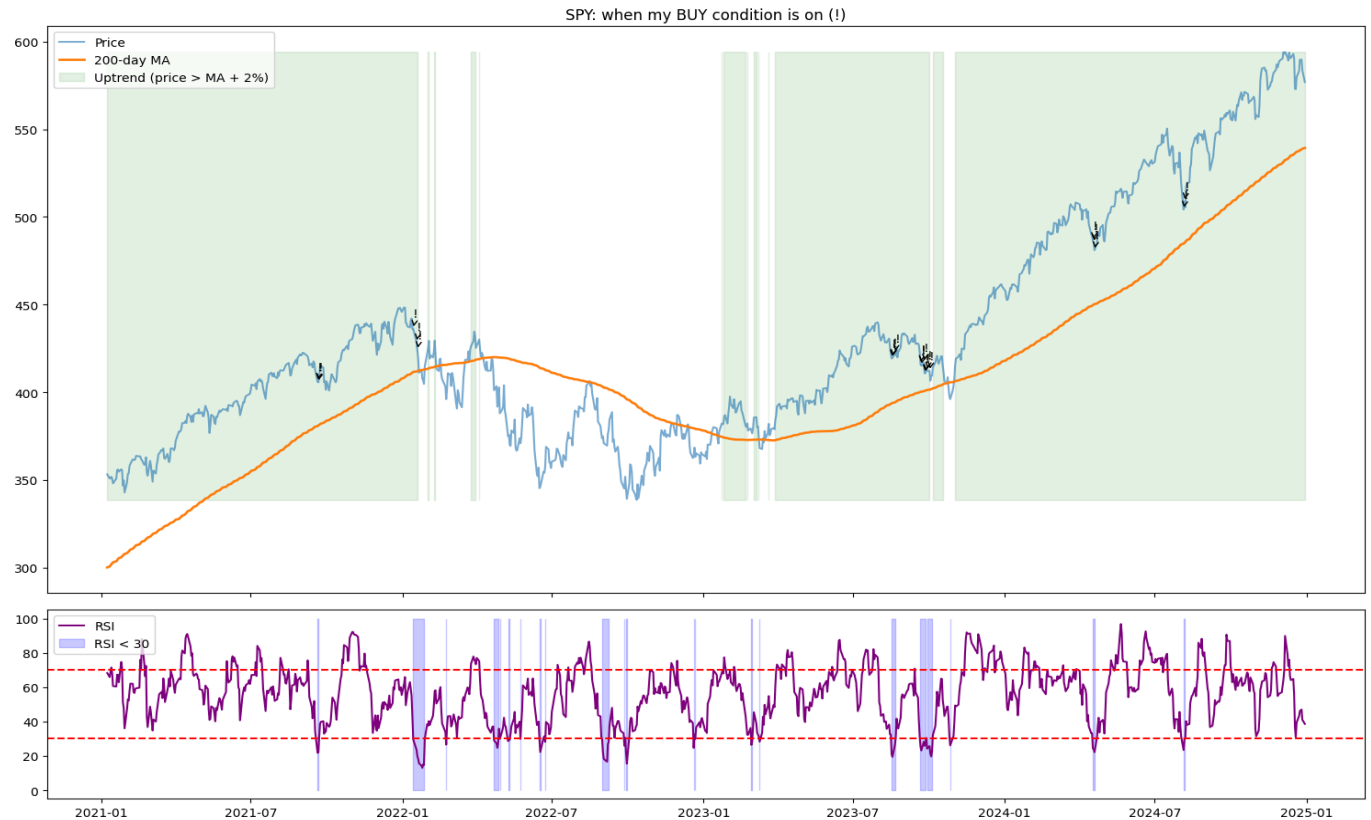
7. Putting It Together: My Confluence Strategy (MA + RSI)

Finally, I combined the two ideas into my own strategy. The 200-day MA tells me the long-term trend. RSI tells me when there's a short-term dip.

- **Buy** when the price is above the 200-day MA (+2%) **and** RSI is below 30. In other words, buy dips, but only in an uptrend.
- **Sell** when the price is below the 200-day MA (-2%) **and** RSI is above 70.

My first version only fired on the exact day a level was crossed. I changed it to check *states* instead: both conditions have to be true on the same day. While looking at the Confluence trades, I noticed a pattern. Some trades sold too early, and others gave back a big part of their gains before the sell signal came. My guess was that an ATR stop could fix this, by protecting gains that had already been made. So I tested a wider 3.5x ATR stop on Confluence. It didn't really help. Since it added complexity without adding value, I dropped it and kept the strategy simple.

Confluence ended up with the best Sharpe ratio (0.66), a max drawdown of about -34%, and a win rate of about 88%. That win rate comes from only 32 trades in 20 years, though, so one or two trades can move it a lot. It still didn't beat buy & hold on total return, but it came the closest.



Evaluating Risk vs. Reward

To decide whether any of this was worth the effort compared to just holding, I looked at total return, maximum drawdown (the biggest drop from a peak), and the Sharpe ratio:

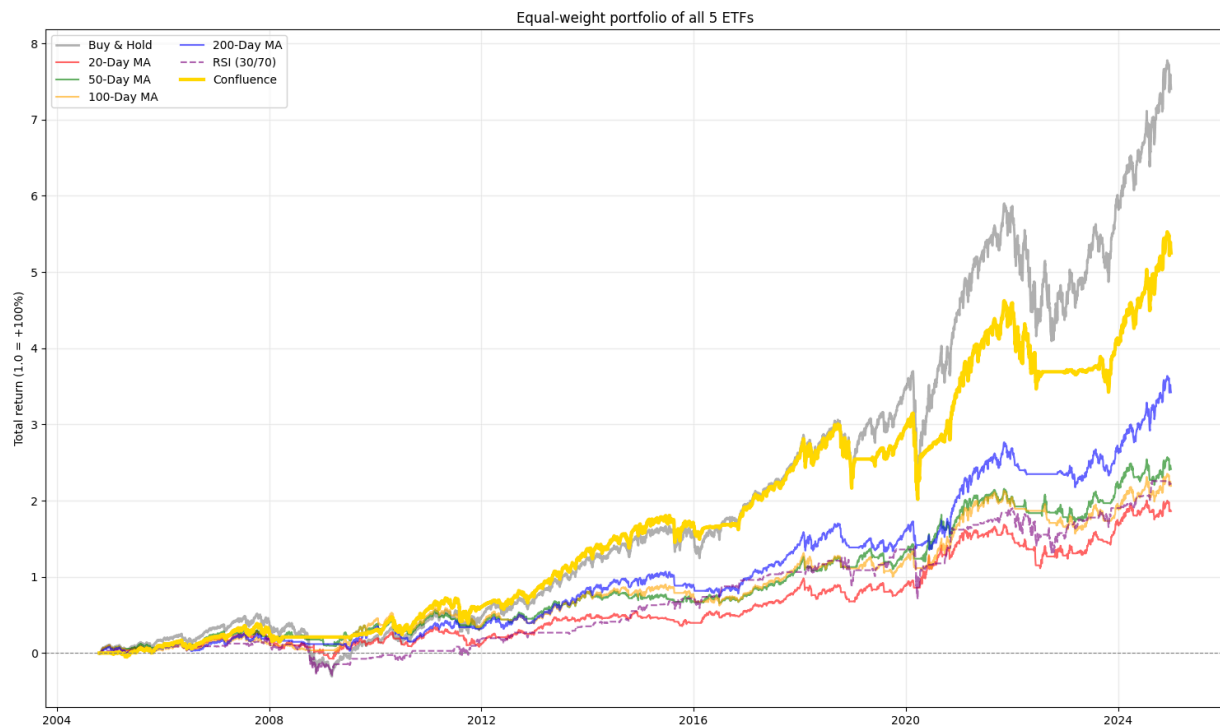
Sharpe ratio = (average daily return – risk-free rate) ÷ standard deviation of daily returns, × $\sqrt{252}$

I used 0% for the risk-free rate. The table shows the average across the 5 ETFs.

Strategy	Total Return	Max Drawdown	Sharpe Ratio	Win Rate	Trades (all 5 ETFs)
Buy & Hold	791%	-54.9%	0.62	—	5
Confluence (MA + RSI)	548%	-34.3%	0.66	87.6%	32
200-Day MA	354%	-23.8%	0.64	52.5%	307

Strategy	Total Return	Max Drawdown	Sharpe Ratio	Win Rate	Trades (all 5 ETFs)
50-Day MA	253%	-24.4%	0.56	51.2%	315
100-Day MA	225%	-26.0%	0.53	51.0%	305
RSI (30/70)	215%	-45.0%	0.43	81.3%	270
20-Day MA	199%	-31.7%	0.48	49.5%	422

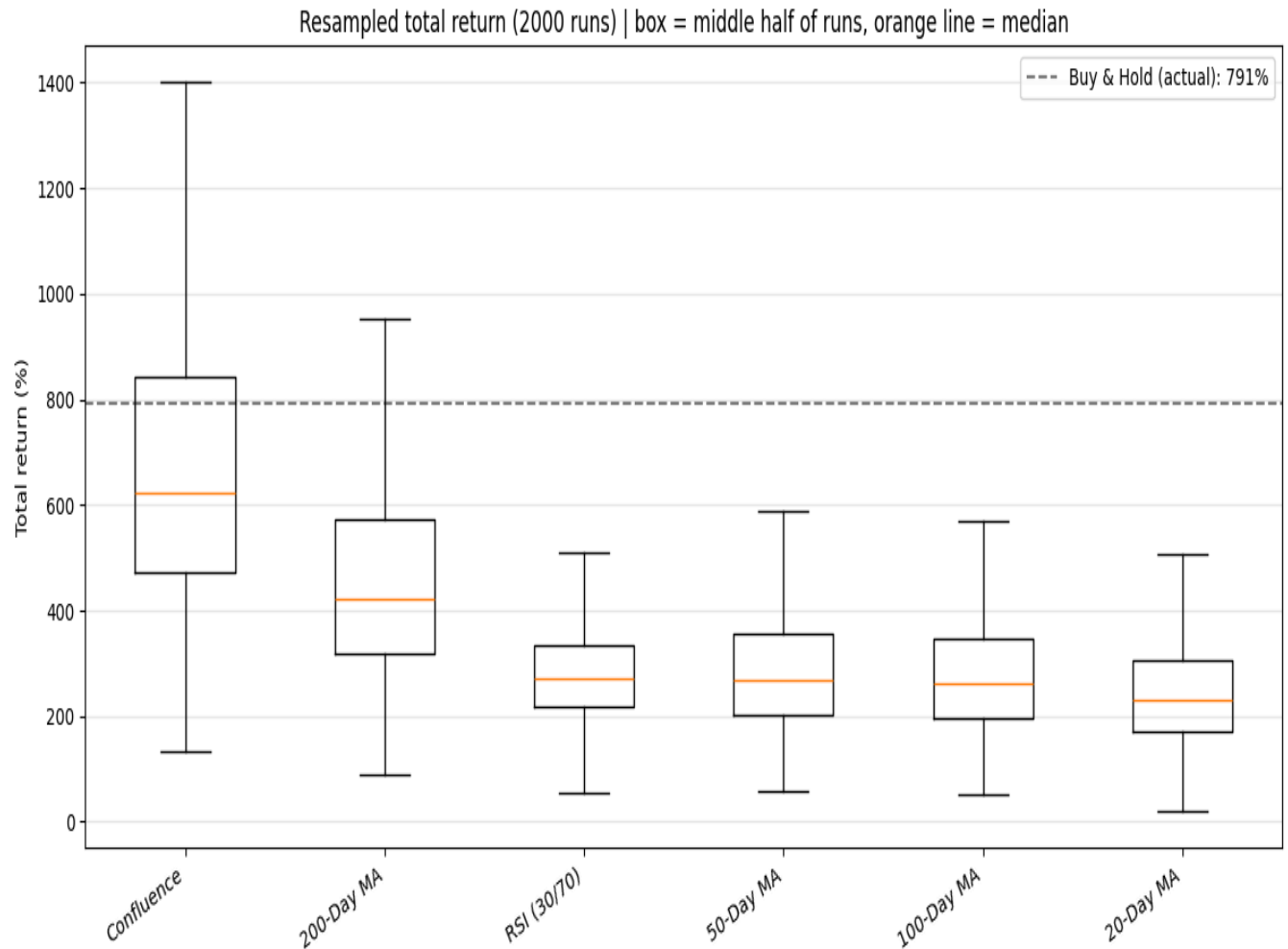
Buy & hold won on return by a mile. The 200-day MA had the smallest crashes. Confluence and the 200-day MA had slightly better Sharpe ratios than buy & hold, but only slightly. QQQ had the biggest returns for every strategy except RSI, and it pulls the averages up. IWM (small caps) was usually the smallest.



The chart surprised me. From about 2009 to 2019, my Confluence line sits almost right on top of buy & hold. That's because its sell rule hardly ever fires, so it basically turned into buy & hold. It was different in only a few places: it was out of the market in 2008 (which helped), it rode the 2020 COVID crash all the way down, and it sat in cash for part of 2022–23 and missed the start of the rebound.

Did I Just Get Lucky?

Confluence made only about 6 trades per ETF. Maybe it just got lucky with a few big ones? To check, I took each strategy's own list of trades, randomly picked the same number of trades from it (the same trade could be picked twice), and compounded them. I did this 2,000 times per strategy. This treats every trade as independent of the others, which is a simplification.



Confluence has the tallest box by far, so its result depends the most on which trades you happen to get. Its middle run (622%) is higher than the 200-day MA's (423%). Confluence may be better, but the boxes overlap too much to tell. It beat buy & hold in only about 30% of runs, the 200-day MA in about 8%, and the rest almost never. The middle runs are also a bit higher than the real results. That's because each run shuffles each ETF's trades separately and then averages the 5 ETFs. A lucky run on one ETF (especially QQQ) can shoot way up, while losses can't go below -100%, so averaging pulls the middle up.

I also re-ran everything with lots of other settings: different stop sizes, buffer sizes, and Confluence MA lengths and RSI levels. The main story held every time. One thing did surprise me: **Confluence only works with a long moving average.** With a 200-day MA it made 548%, with a 150-day MA 498%, with a 100-day MA 236%, and with a 50-day MA just 4% (one trade in 20 years). With a short MA, a dip big enough to push RSI below 30 usually also pushes the price below the MA, so the buy rule almost never fires.

Conclusion

My main question was whether momentum could beat buy & hold. Here's how my three ideas held up:

1. **Momentum reduces big drawdowns: yes.** The 200-day MA cut the worst drop from about -55% to about -24%. But it ended with less than half the money.
2. **RSI depends on volatility: partly.** RSI did best on IWM (small caps, which bounce around more). But it was weakest on DIA (the Dow), so it's not a clean pattern. With only 5 ETFs, this is weak evidence anyway.
3. **More complexity means more consistent returns: not really.** Some added rules helped (the buffer, combining trend and RSI). Others hurt (tight stops). My most complex strategy, Confluence, has the best Sharpe ratio, but it also depends the most on luck. At first I really liked its high win rate because it felt smoother and more comfortable. Later I realized a high win rate can be misleading if it comes with lower long-term returns or missing big market moves.

Bogle wins on growth. Passive indexing beat every strategy I built over 20 years. The price is a painful -55% drop along the way. Trading rules don't add return; what they can do is control risk and protect money during a crash. When I start investing, I'll go with buy & hold. If I couldn't handle a -55% drop, the 200-day MA is the one I'd consider, knowing I'd end up with a lot less money.

My Takeaways

This project changed how I think about investing and about building models. I expected more sophisticated trading rules to keep beating passive investing. Instead, every bit of complexity brought new trade-offs, and long-term compounding stayed very hard to beat. I also learned to let go of ideas that don't work. I prototyped RSI levels on short 5-year windows and tried an ATR stop on Confluence. Neither held up in the full test, so I removed them.

I also learned how easily a backtest can fool you. Some of my best-looking results came from bugs, not from good strategies. Now I don't trust a number until I've checked it a second way.

Limitations & What I'd Try Next

This backtest taught me a lot, but it has some real limits:

- **No trading costs.** The code assumes perfect trades with no commissions, no bid-ask spread, and no fees. Strategies that trade a lot, like the 20-day MA, would lose even more in real life.
- **Taxes.** Buy & hold gets to delay taxes for years. Active strategies keep triggering short-term capital gains taxes, which would drag down real returns.
- **Parameter choices and overfitting.** I picked settings like the 2% buffer and 5% stop after testing by hand, so there's a risk they just happen to fit these 20 years. Testing lots of other settings showed the main story doesn't change, but Confluence does depend on using a long moving average.
- **One time period.** 2004–2024 included a long stretch of falling interest rates and easy money, which favored being in the market.
- **Only five ETFs.** All five are big, liquid US funds, and they move together a lot. The results might not apply to individual stocks, which are much riskier.
- **Long-only.** The strategies can't short-sell. In a bear market they can only move to cash (earning 0%), not profit from prices falling.

Technical Challenges & Iteration

Sharpe ratios that were way too high

Early on, some strategies showed Sharpe ratios above 10, which is not realistic. When I dug in, I found my mistake. I was calculating the ratio from a short list of trade returns instead of a daily return series. Multiplying by $\sqrt{252}$ only makes sense for daily data, and my version also ignored all the days the strategy sat in cash earning 0%. I wrote a function that turns each list of trades into a daily return series, with 0% on cash days. After that, the Sharpe ratios came back to realistic levels (around 0.6).

Charts and tables that disagreed

My growth charts showed about 200% growth, while my tables showed about 390% for the same strategy. The charts were adding returns together, while the tables compounded them. Compounding is the correct way, because each gain builds on the last one. I switched the charts to compound too (using `cumprod`), and then the charts and tables matched.

The ATR stop that traded too much

My first ATR stop made about 300% more trades than the fixed stop, and all those extra trades hurt returns. To figure out why, I measured how far below the price each stop sat. The 2x ATR stop was only about 2.7% away on average, compared to 5% for the fixed stop. It was simply too tight. Wider ATR stops (like 3.5x) and the fixed 5% stop basically tied, so I kept the simpler one.

Making the start dates fair

Buy & hold started on day one, but the 200-day MA can't give a signal until it has 200 days of data. That wasn't a fair comparison. So I made every strategy in the main table skip the first 200 days. Now they're all measured over exactly the same period.

Cleanup and final review

For the final version, I used Gemini to help with debugging and editing. I also made a few small fixes:

- **Trade timing.** Trades happened at the same closing price as the signal, but you can't actually trade at a price you only see after the close. Now every trade happens at the next day's close. This is the fix that reshuffled my ATR ranking.
- **Open trades at the end.** Trades that were still open on the last day weren't being counted.
- **Selling shares I didn't own.** Confluence could send a "sell" signal before it had ever bought. Now it can only sell after buying.
- **Leftover adding.** Some older tables and charts still added returns instead of compounding them.

Some numbers moved, but the main result didn't change: buy & hold still came out ahead. It was one more reminder to always double-check your code.

Versions

v1 (spring 2026): original analysis. After fixing the cash-day bug, the 200-day MA's Sharpe ratio was 0.627 (buy & hold 0.619).

v2 (September 2026): small fixes to trade timing and bookkeeping. Some numbers moved (200-day MA Sharpe 0.64, buy & hold 0.62), but the main conclusion didn't change.